

Scenario-normalized objective

Algorithm (mean rank)

RFOTO-ABC (1.15)	1.00	1.00	1.00	1.00	1.00	1.00
LSHADE-lite (4.26)	1.08	1.05	1.02	1.02	1.01	1.01
JADE (4.42)	1.12	1.04	1.02	1.02	1.01	1.01
Gbest-ABC (4.52)	1.13	1.05	1.02	1.02	1.01	1.01
MeABC (4.59)	1.13	1.05	1.02	1.02	1.01	1.01
GWO (4.66)	1.14	1.05	1.02	1.02	1.01	1.01
CMAES-lite (4.66)	1.14	1.05	1.02	1.02	1.01	1.01
GA (7.75)	1.31	1.79	1.71	1.94	1.84	1.79
Standard-ABC (9.00)	2.22	2.58	2.26	2.36	2.19	2.07
PSO (10.00)	3.33	3.78	3.01	2.67	2.75	2.30
DE (11.00)	4.42	5.37	4.73	3.23	3.54	2.90
	S1	S2	S3	S4	S5	S6

Mean objective / scenario best